NCBEE 2026

Age-norms across cultures: effects on team decision-making

Nhi Cao
nhi.cao@hanken.fi

Hanken School of Economics
Helsinki Graduate School of Economics, Finland

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Motivation

Social norms in economics

Social norms proscribe behaviour (Krupka & Weber, 2013)

- Injunctive norms, what people collectively think is appropriate to do
- Descriptive norms, what people typically do in a situation

People might follow norms

- *intrinsically* (cognitive dissonance when not)
- to gain *social approval* (reputational loss or social punishment if not)
  - degree of social influence

"behaviour sustained by expectations and informal sanctions" (Bicchieri, 2005)

What do we know about norms?

1. People have intrinsic beliefs about what's appropriate and behave accordingly (Krupka & Weber, 2013)

2. Norms differ across societies and cultures (Herrmann et al., 2008; Gächter & Herrmann, 2009; Almås et al., 2025)
   - Some norms are more 'social' than others, depending on how many people share the norms (Bicchieri, 2005)
   - Some norms are observed in some cultures and not others.

Confucianism in Asian culture

Confucian values and beliefs in East Asian countries:

- respect parent, elders, ancestors
- older people are more knowledgeable

Translate to behavioural norms:

- address an older person with age-appropriate pronouns
- live with and care for parents in old age
- promote older candidates for decision-making positions
- do not challenge an older person

$\Rightarrow$ some might be relevant for development, but so far understudied.

I compare: Vietnamese and American teams in incentivized group tasks where the leader differs in age
$\Rightarrow$ are VN teams less likely to challenge an older leader (than US teams)?

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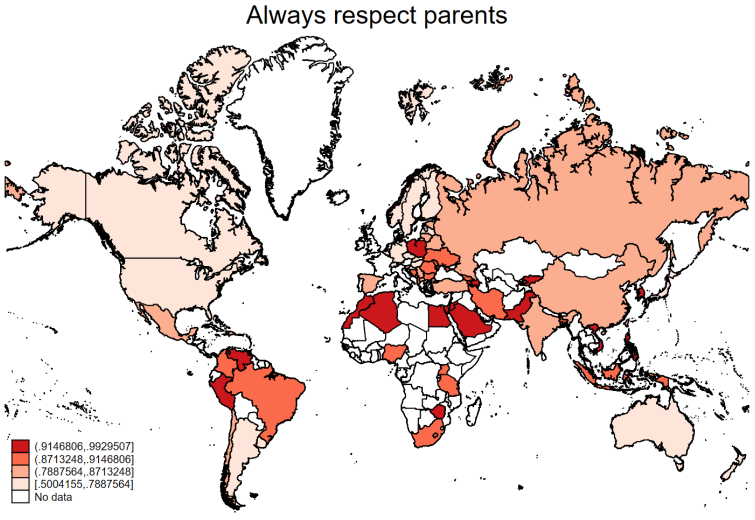
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Motivating evidence



Percent responds 'Yes'

(Source: World Value Survey 1984-2020, every 4 years.)

Research question

Do people follow age-norms and not challenge an old leader? Even if there is incentive to do the opposite and object?

Run an experiment focused on the **VN-US difference** in the behavioural norm

... where people ***do not challenge an older person***

... in team-working context where the **team leader is an older person working with younger subordinates**.

→ Why this norm? Affects team productivity, feedback and learning, creative destruction, ... could matter for development

Literature

Cultural economics

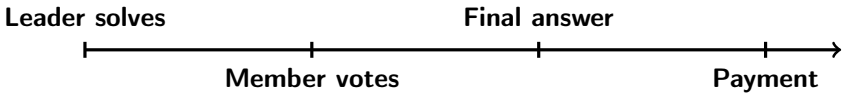
- living arrangement, language, gender roles, ... (Lowes, 2020; Bau, 2021; Jakiela & Ozier, 2020; Squires, 2024)
- rule-following (Fisman & Miguel, 2007; Gneezy & Rustichini, 2000)
- cultural divergence (Mokyr & Tabellini, 2024; Jia & Kung, 2025; Moscona et al., 2026), universalism (Enke, 2019)
- Norms are sometimes misperceived (Bursztyn et al., 2020) ...
- ... and can be triggered (Shih et al., 1999)

Behavioural economics

- leader's status and identity (Gangadharan et al., 2025)
- leader identity and coordination (Bhalotra et al., 2023)

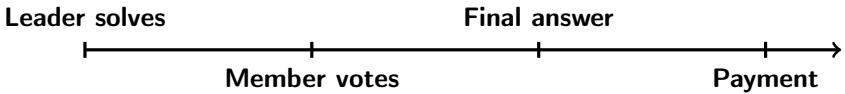
Experiment

The game



[1] Group size and MDE bench-marked from Bessey (2023) and Curşeu et al. (2013) MDE

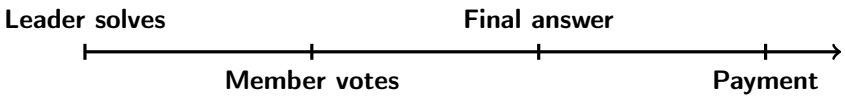
The game



- Teams of 3 members[1] and 1 leader solve 4 Raven matrix quizzes and get paid per correct answer

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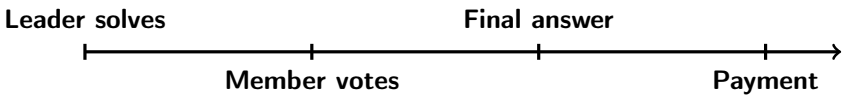
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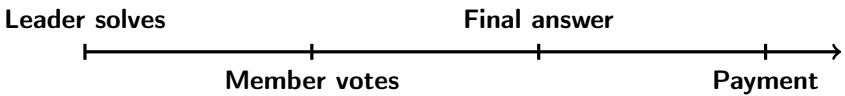
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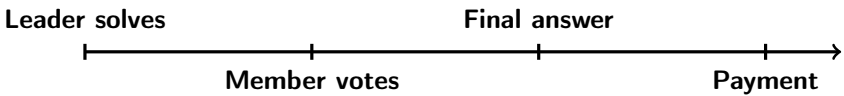
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- If majority rejects leader's answer, a random one is picked from remaining options and team is rewarded if the pick is correct (otw. not).

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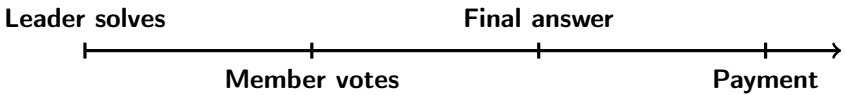
The game



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- Leader attempts the quizzes first, then members vote to accept/reject (sequential).
- Decision is made by majority rule. If final answer is correct, team is rewarded (otw. not).
- If majority rejects leader's answer, a random one is picked from remaining options and team is rewarded if the pick is correct (otw. not).
- Randomly pick 1 out of 4 quizzes/questions to check for reward.

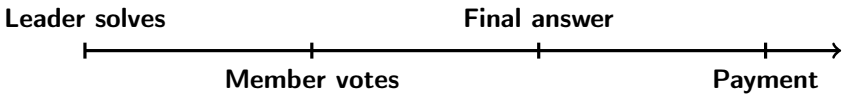
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The game (cont.)



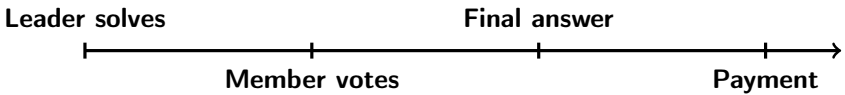
- Quizzes are in 2 difficulty levels: first one easy, next three hard. Same sequence for everyone.

The game (cont.)



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- 1 quiz/question = 1 round/decision (4 in total), partners-matching (no rematching between rounds)

The game (cont.)



- Quizzes are in 2 difficulty levels: first one easy, next three hard. Same sequence for everyone.
- 1 quiz/question = 1 round/decision (4 in total), partners-matching (no rematching between rounds)
- Members do not learn about others' behaviour, no info revealed between rounds, no incentive to cooperate

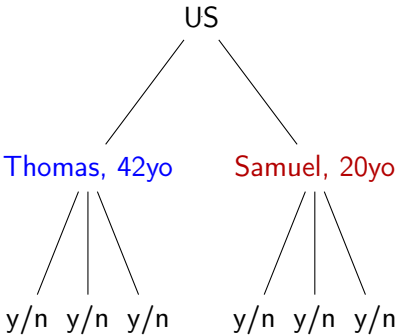
Treatment variation

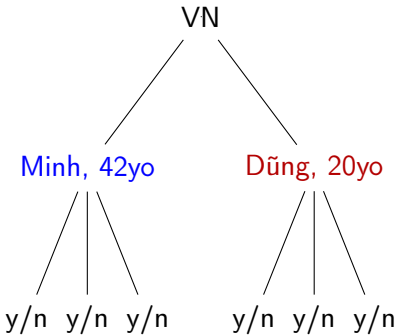
Each team is assigned randomly to either an **Old leader** or a **Young leader**

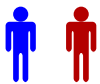
- 2 leaders recruited: 42yo (Old leader) and 20yo (Young leader) males
- Leaders have dual names/ethnic identities: 1 Vietnamese name, 1 English name. For VN subject pool, Vietnamese names are shown; for US pool, English names are shown.

The experiment is done on 2 homogenous cultural subject pools: Vietnamese (VN) and Americans (US).

Illustration







old leader
young leader

Recruitment - Prolific platform

Sample restrictions for each cultural subject pool:

- nationality, residence, mother tongue, raised mono-lingual

In-study screening:

- attention check for instruction comprehension and effort
- language and culture screening questions for VN subject pool

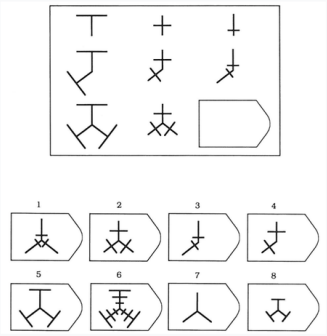
Snapshot: Leader view

Question 1

There are eight figures going from left to right, top to bottom.

They follow a pattern, and there is one missing.

Choose from options 1-8 the figure that completes the pattern. What goes in the blank?



Your answer:

○ 1 ○ 2 ○ 3 ○ 4 ○ 5 ○ 6 ○ 7 ○ 8

Next

Snapshot: Member view, between-subject

Treatment ***Young***

The Leader - Samuel, aged 20 - proposed:

figure 1

Do you accept Samuel's (20yr) answer as your group's final answer?

- ○ Yes
- ○ No

Next

Treatment ***Old***

The Leader - Thomas, aged 42 - proposed:

figure 2

Do you accept Thomas' (42yr) answer as your group's final answer?

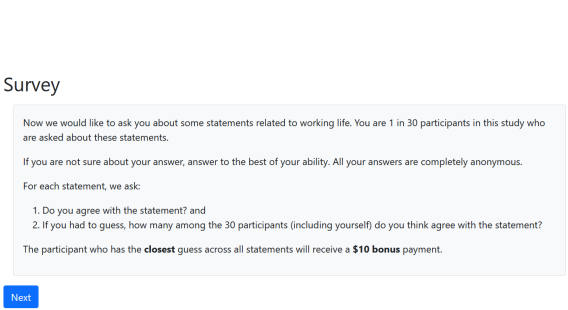
- ○ Yes
- ○ No

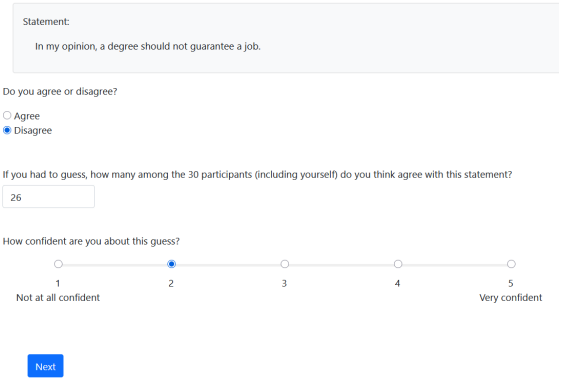
Next

Snapshot: Member post survey

Krupka-Weber survey on social appropriateness on **two age-related statements** (+ masking statements)

- In a group, **people should rely on an older person** rather than a younger one to make decisions.
- A **40-year-old boss is more acceptable** than a 20-year-old one.

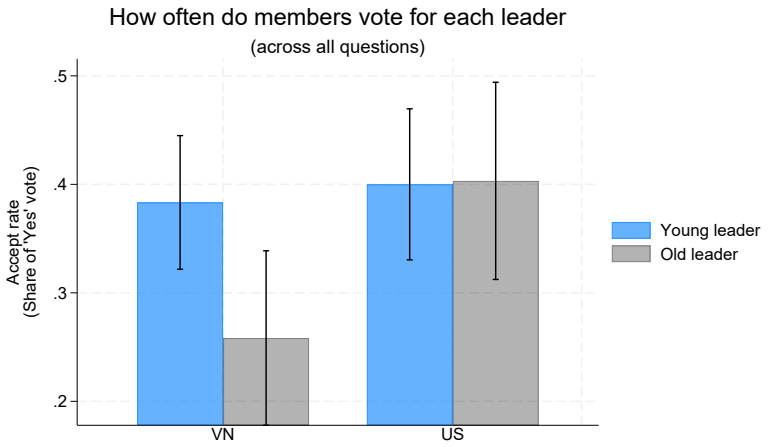




Accept rate

Do VN teams endorse older leader more? (less likely to reject)

Accept rate


How often do members vote for each leader
(across all questions)
Accept rate
(Share of 'Yes' vote)
.5
.4
.3
.2
VN
US
Young leader
Old leader

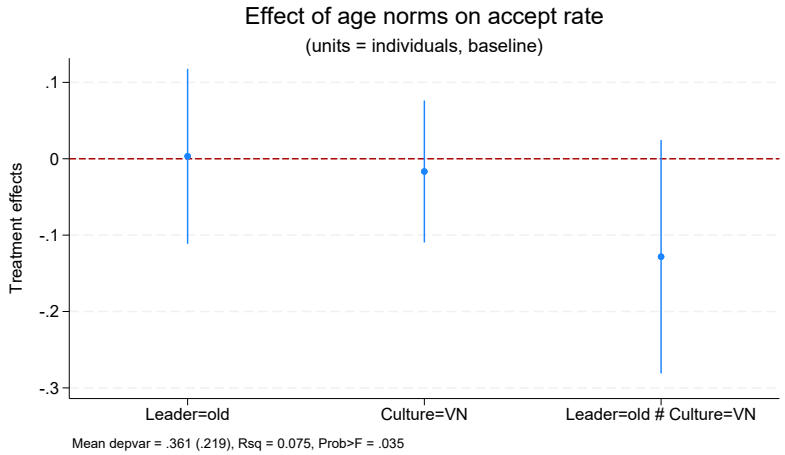
Empirical model

$$y_i = \beta Old_i + \gamma VN_i + \boldsymbol{\delta Old_i \cdot VN_i} + X'\psi + \epsilon_i \quad (1)$$

- $Asian_i$ =1 if participant i is from VN subject pool, =0 if US
- Old_i =1 if participant i is assigned to the old leader, =0 if young leader
- X vector of demographic controls
- ϵ_i error term
- y_i outcomes: no of votes (count), accept rate (*yes* votes of total questions), no of correct votes (count), accuracy rate (correct votes out of total questions)

$\Rightarrow$ Expect $\delta > 0$, i.e., should see VN teams endorse Old leader more, relative to US teams.

Accept rate


Effect of age norms on accept rate
(units = individuals, baseline)
Treatment effects
.1
0
-.1
-.2
-.3
Leader=old
Culture=VN
Leader=old # Culture=VN
Mean depvar = .361 (.219), Rsq = 0.075, Prob>F = .035

Accept rate

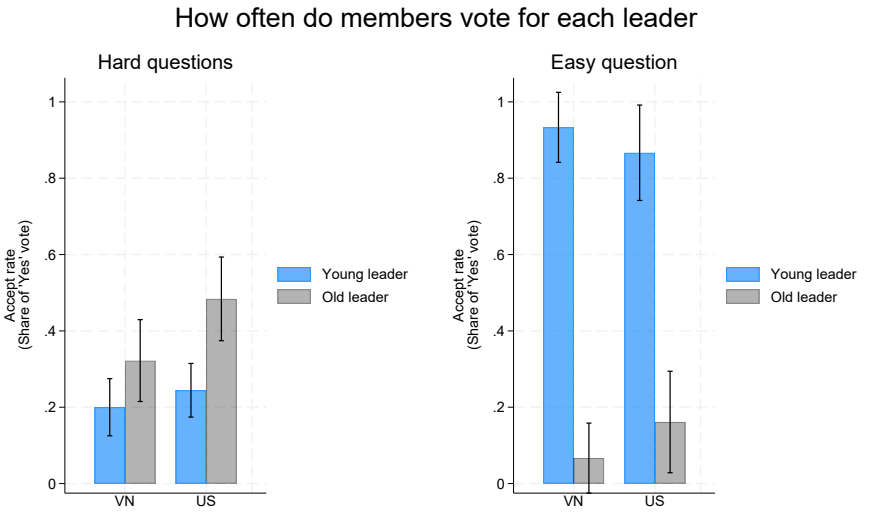
	Baseline	Spec 1	Spec 2
Leader=old	0.003	-0.051	-0.063
	(0.06)	(0.07)	(0.07)
Culture=VN	-0.017	-0.050	-0.064
	(0.05)	(0.05)	(0.06)
Leader=old × Culture=VN	-0.128*	-0.059	-0.040
	(0.08)	(0.09)	(0.09)
Age		0.003	0.003
		(0.00)	(0.00)
=1 if female		0.033	0.027
		(0.04)	(0.04)
=1 if student		0.138**	0.131*
		(0.07)	(0.07)
=1 if employed		0.016	0.026
		(0.04)	(0.05)
Completion time			-0.003
			(0.00)
Total approvals			-0.000
			(0.00)
Mean(depvar)	0.362	0.363	0.363
Prob > F	0.0345	0.0080	0.0098
N	121	106	106
Demographic controls	No	Yes	Yes
Heuristic controls	No	No	Yes

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Does this pattern change under uncertainty?
Are people more likely to endorse or challenge the old leader when they don't know the answer?

Uncertainty

Accept rate under uncertainty



Accept rate under uncertainty

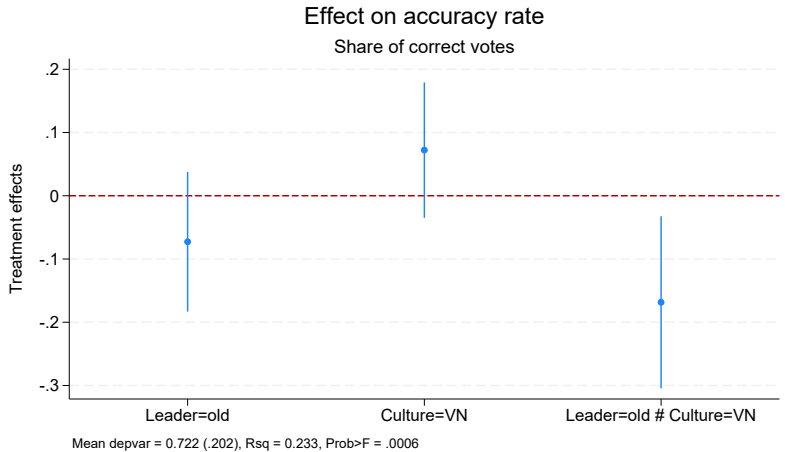
	Accept (all)	Accept (hard)	Accuracy (all)	Accuracy (hard)
Leader=old	-0.063	0.163**	-0.073	-0.078
	(0.07)	(0.08)	(0.06)	(0.06)
Culture=VN	-0.064	-0.088	0.072	0.076
	(0.06)	(0.07)	(0.05)	(0.06)
Leader=old × Culture=VN	-0.040	-0.015	-0.168**	-0.238***
	(0.09)	(0.11)	(0.07)	(0.09)
Mean(depvar)	0.363	0.314	0.722	0.660
Prob > F	0.0098	0.0029	0.0006	0.0001
N	106	106	106	106
Demographic controls	Yes	Yes	Yes	Yes
Heuristic controls	Yes	Yes	Yes	Yes

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

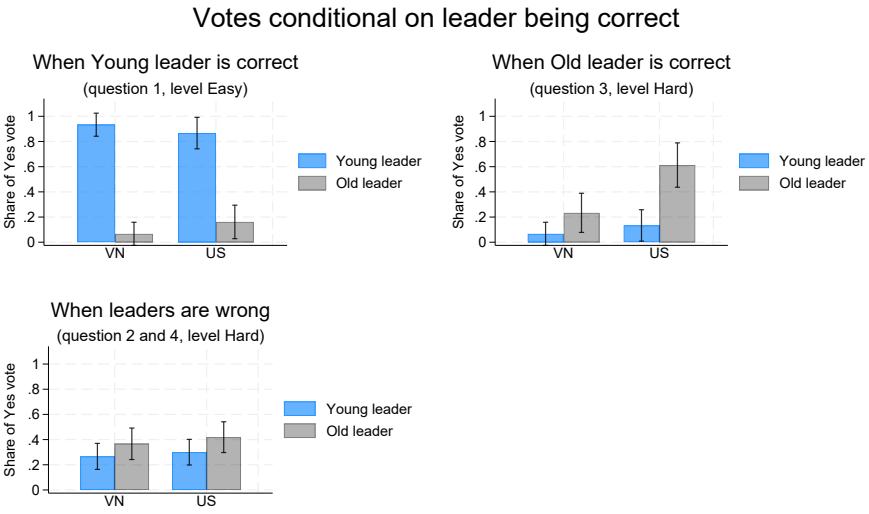
What about performance? Were VN groups correct in rejecting old leader's answer?

Accuracy rate (share of correct votes)

Correct = 1 if (`accept` and leader's `correct`) or (`reject` and leader's `incorrect`)



False negatives: reject when leader's correct



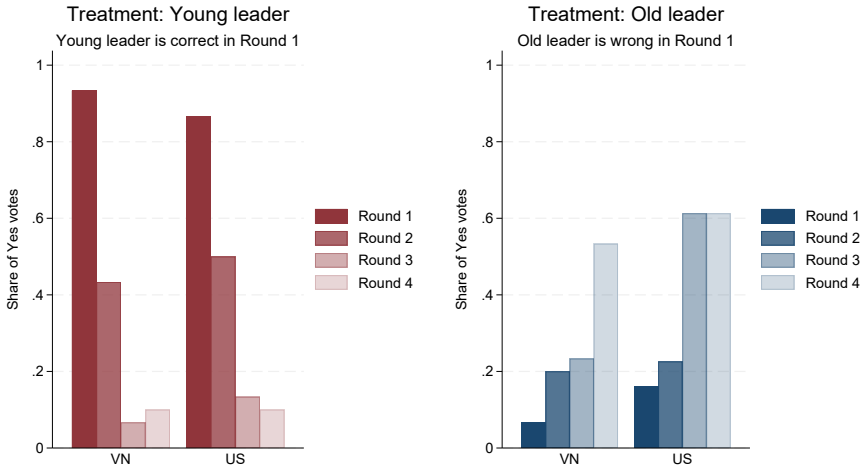
Accuracy rate under uncertainty

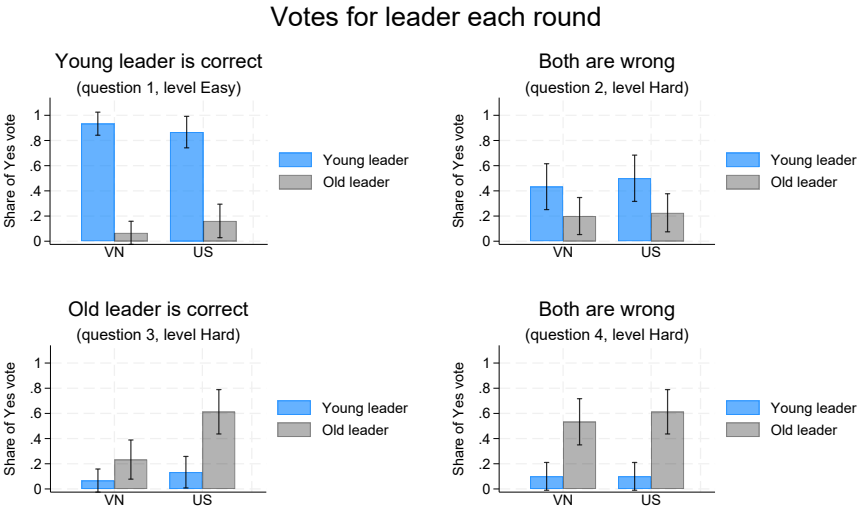
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	(0.07)	(0.08)	(0.06)	(0.06)
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N	106	106	106	106
Demographic controls	Yes	Yes	Yes	Yes
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* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Learning

How members vote after learning about leader's skills




Votes for leader each round
Young leader is correct
(question 1, level Easy)
Both are wrong
(question 2, level Hard)
Old leader is correct
(question 3, level Hard)
Both are wrong
(question 4, level Hard)
Share of Yes vote
0
.2
.4
.6
.8
1
VN
US
Young leader
Old leader

Summing up

1. Asians are **more likely (than Westerners) to challenge** an older leader (imprecise, unexpected).
2. Age-norms **kick in under uncertainty** for **both cultural groups**. People are less likely to challenge the older leader (rel. to younger leader) when they don't know the answer. When people are certain about the true answer, age-norms are no longer salient.
3. Asians are **less accurate** because they often reject when the old leader is right.

Take-away

Where does this leave us? Don't know yet. Need to understand more.

For main collection:

- Incentive structure that does not punish voting No and elicits cost of challenging leader
- Treatment arm no. 2: leader's cultural identity
- Treatment arm no. 3: leader's merits e.g. university degree, record of solved problems, ...

Your comments are appreciated!

nhi.cao@hanken.fi

Appendix

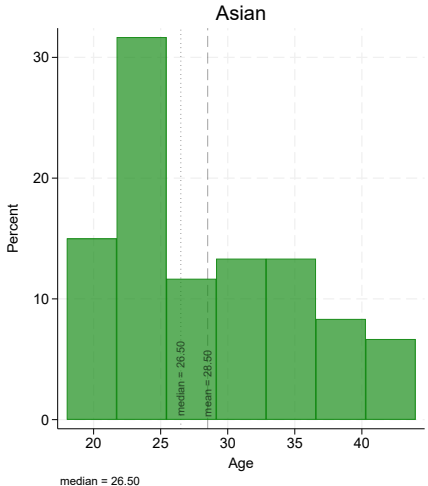
Belief survey and heterogeneity analysis

Belief elicitation: survey results

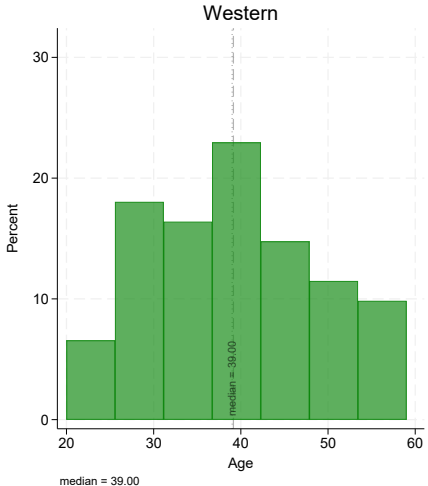
Heterogeneity in voter's age: age groups

Accept rate conditional on belief: conditional accept rate

Age distribution of voters



median = 26.50
mean = 28.50



median = 39.00
mean = 39.15

back to Appendix

Subsample: above and below certain age

	Baseline	Base, voters <= 30yo	Base, voters > 30yo	Base, voters <= 27yo	Base, voters > 27yo
Leader=old	0.013	0.029	0.007	0.083	-0.011
	(0.23)	(0.59)	(0.26)	(0.79)	(0.25)
Culture=asian	-0.067	0.008	-0.154	0.283	-0.121
	(0.19)	(0.26)	(0.23)	(0.30)	(0.23)
Leader=old × Culture=asian	-0.513*	-0.608	-0.370	-0.499	-0.599
	(0.31)	(0.65)	(0.43)	(0.84)	(0.41)
Mean(depvar)	1.446	1.360	1.507	1.308	1.512
Prob > F	0.0345	0.1120	0.4681	0.5108	0.1446
N	121	50	71	39	82
Demographic controls	No	No	No	No	No
Heuristic controls	No	No	No	No	No

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

back to Appendix

Age groups as treatment interaction

	Baseline	Spec1	Spec2
Leader=old	0.013	0.029	0.083
	(0.23)	(0.59)	(0.78)
Culture=VN	-0.067	0.008	0.283
	(0.19)	(0.26)	(0.29)
Leader=old × Culture=VN	-0.513*	-0.608	-0.499
	(0.31)	(0.64)	(0.82)
above 30yo		0.037	
		(0.26)	
Leader=old × above 30yo		-0.022	
		(0.64)	
Leader=old × Culture=VN × above 30yo		0.237	
		(0.78)	
above 27yo			0.404
			(0.27)
Leader=old × above 27yo			-0.094
			(0.82)
Leader=old × Culture=VN × above 27yo			-0.100
			(0.92)
Mean(depvar)	1.446	1.446	1.446
Prob > F	0.0345	0.2284	0.1801
N	121	121	121
Demographic controls	No	No	No
Heuristic controls	No	No	No

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

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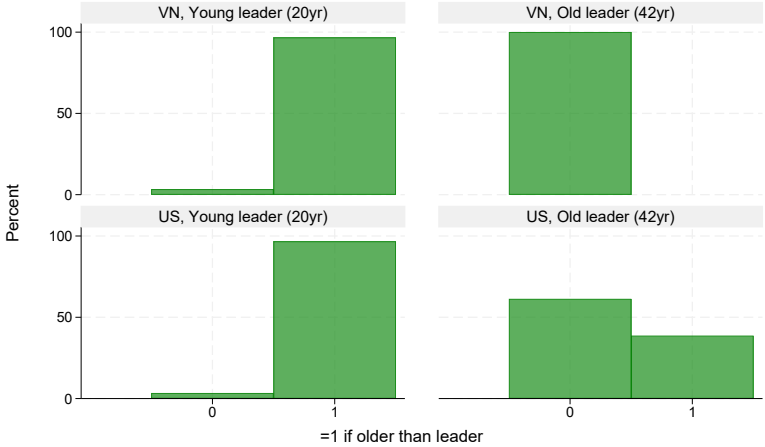
Covariate balance

Covariate balance between western/asian subject pools

	VN			US			Diff.
	N	Mean	Sd	N	Mean	Sd	
Age	60	28.50	6.92	61	39.15	9.93	-10.648***
							(1.554)
=1 if female	60	0.45	0.50	61	0.49	0.50	-0.042
							(0.091)
=1 if student	60	0.22	0.42	49	0.12	0.33	0.094
							(0.072)
Employment	60	0.60	0.49	50	0.66	0.48	-0.060
							(0.093)
Completion time	60	16.17	9.37	61	10.66	4.92	5.506***
							(1.364)

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Heterogeneity: How many are older than leader?



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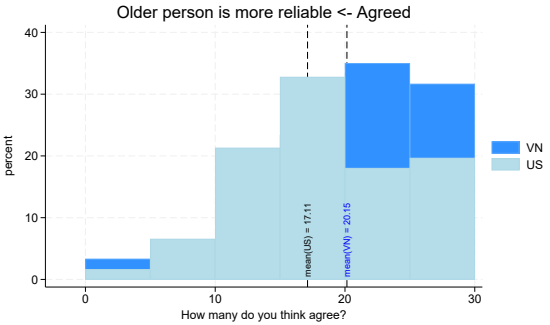
Subsample: younger/older than leader

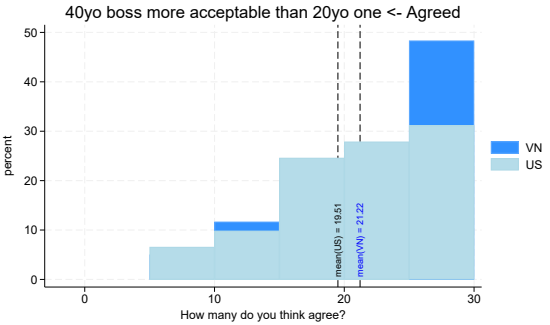
	Baseline	VN, voter < leader	US, voter < leader	US, voter > leader
Leader=old	0.013			
	(0.23)			
Leader=old		-0.967***	0.421	0.296
		(0.17)	(0.28)	(0.24)
Culture=asian	-0.067			
	(0.19)			
Leader=old × Culture=asian	-0.513*			
	(0.31)			
Mean(depvar)	1.446	1.065	1.400	1.707
Prob > F	0.0345	0.0000	0.1436	0.2222
N	121	31	20	41
Demographic controls	No	No	No	No
Heuristic controls	No	No	No	No

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

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Survey of beliefs

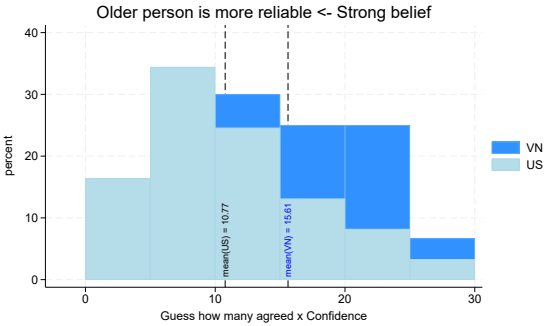


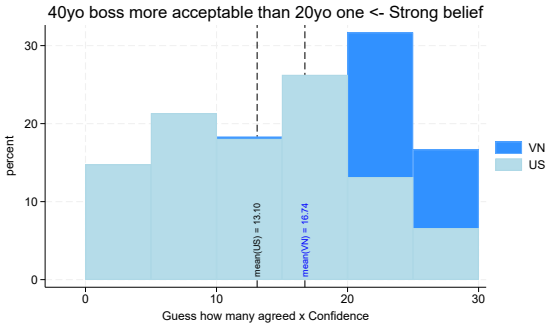


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How strong are beliefs

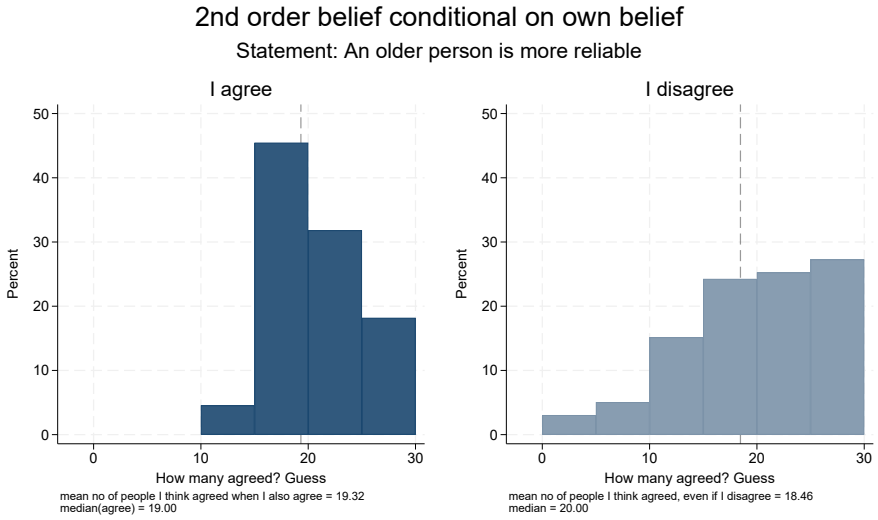
Guesses are scaled by confidence rate from .25 to 1 (original survey scale 1-5)





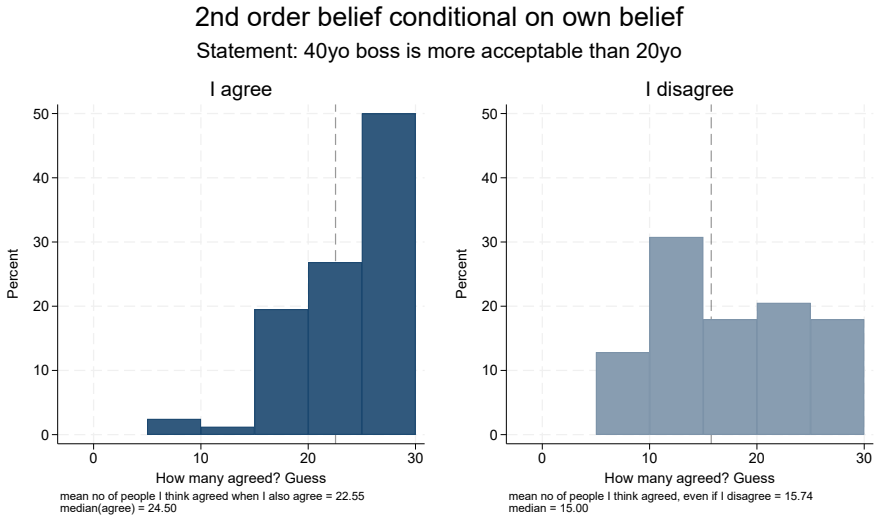
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2nd order belief 1/



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2nd order belief 2/



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Heterogeneity: vote count conditional on belief

Conditional on Disagree to "Older people are more reliable."

	Baseline	Base, Disagree	Spec2, Disgree
Leader=old	0.013	-0.700	-0.925
	(0.23)	(0.44)	(0.57)
Culture=VN	-0.067	-0.800***	-1.058***
	(0.19)	(0.29)	(0.38)
Leader=old × Culture=VN	-0.513*	0.300	0.706
	(0.31)	(0.56)	(0.65)
Leader=old × Culture=VN × Disagree		-0.978	-1.080
		(0.66)	(0.76)
Mean(depvar)	1.446	1.446	1.453
Prob > F	0.0345	0.0007	0.0095
N	121	121	106
Demographic controls	No	No	Yes
Heuristic controls	No	No	Yes

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

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Dissidents who think they're the minority

Conditional on Disagree to "Older people are more reliable."

	Baseline	Base, Disagree	Base, Disagree alone	Spec2, Disagree alone
Leader=old	0.013	-0.700	-0.500	-0.398
	(0.23)	(0.44)	(0.60)	(0.84)
Culture=VN	-0.067	-0.800***	-0.667**	-0.901**
	(0.19)	(0.29)	(0.29)	(0.39)
Leader=old × Culture=VN	-0.513*	0.300	0.417	0.459
	(0.31)	(0.56)	(0.71)	(0.90)
Leader=old × Culture=VN × Disagree		-0.978	-0.941	-0.646
		(0.66)	(0.85)	(1.03)
Leader=old × Culture=VN × Disagree × High guess			-0.547	-1.787
			(1.21)	(1.26)
Mean(depvar)	1.446	1.446	1.446	1.453
Prob > F	0.0345	0.0007	0.0019	0.0063
N	121	121	121	106
Demographic controls	No	No	No	Yes
Heuristic controls	No	No	No	Yes

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

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Heterogeneity: vote count conditional on belief

Conditional on Disagree to "40yo boss is more acceptable than 20yo one."

	Baseline	Base, Disagree	Spec2, Disagree
Leader=old	0.013	0.173	-0.130
	(0.23)	(0.29)	(0.41)
Culture=VN	-0.067	-0.029	-0.313
	(0.19)	(0.24)	(0.40)
Leader=old × Culture=VN	-0.513*	-0.744*	-0.418
	(0.31)	(0.38)	(0.48)
Leader=old × Culture=VN × Disagree		0.768	0.810
		(0.65)	(0.72)
Mean(depvar)	1.446	1.446	1.453
Prob > F	0.0345	0.0963	0.0845
N	121	121	106
Demographic controls	No	No	Yes
Heuristic controls	No	No	Yes

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

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Dissidents who think they're the minority

Conditional on Disagree to "40yo boss is more acceptable than 20yo one."

	Baseline	Base, Disagree	Base, Disagree alone	Spec2, Disagree alone
Leader=old	0.013	0.173	0.200	0.163
	(0.23)	(0.29)	(0.39)	(0.59)
Culture=VN	-0.067	-0.029	-0.133	-0.247
	(0.19)	(0.24)	(0.38)	(0.48)
Leader=old × Culture=VN	-0.513*	-0.744*	-1.367**	-1.139
	(0.31)	(0.38)	(0.58)	(0.73)
Leader=old × Culture=VN × Disagree		0.768	1.343	1.465
		(0.65)	(0.91)	(1.05)
Leader=old × Culture=VN × Disagree × High guess			-0.676	-0.691
			(1.15)	(1.35)
Mean(depvar)	1.446	1.446	1.446	1.453
Prob > F	0.0345	0.0963	0.0028	0.0594
N	121	121	121	106
Demographic controls	No	No	No	Yes
Heuristic controls	No	No	No	Yes

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

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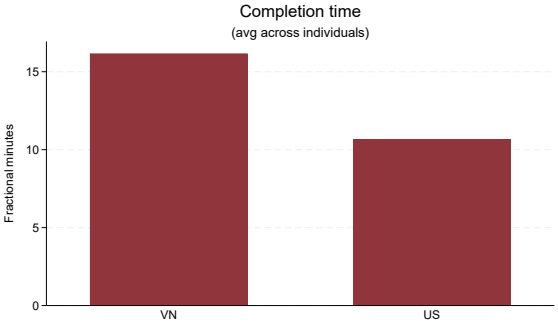
Covariate balance

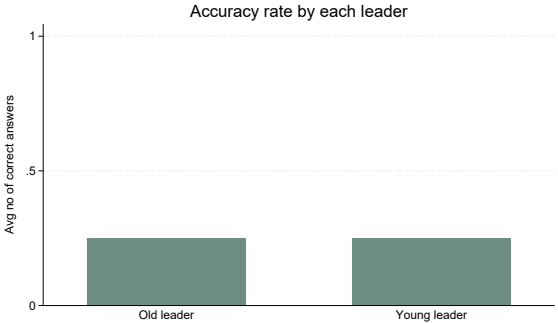
Covariate balance between treatment conditions

	Leader=old			Leader=young			Diff.
	N	Mean	Sd	N	Mean	Sd	
=1 if VN	61	0.49	0.50	60	0.50	0.50	-0.008
							(0.092)
Age	61	34.21	10.26	60	33.52	9.95	0.696
							(1.837)
=1 if female	61	0.52	0.50	60	0.42	0.50	0.108
							(0.091)
=1 if student	57	0.14	0.35	52	0.21	0.41	-0.071
							(0.074)
Employment	56	0.59	0.50	54	0.67	0.48	-0.077
							(0.093)
Completion time	61	13.43	8.75	60	13.35	7.08	0.075
							(1.446)
Total approvals	61	1454.21	2870.35	60	1772.10	2974.38	-317.887
							(531.517)

Employment=1 if full-time or part-time employed, =0 if unemployed or not in workforce.
Total approvals: no of approved submissions on Prolific.

Descriptives





Tabulation of beliefs

	Disagree	Agree
to both	34	17
to one	70	70
to neither	17	34
	121	121

34 dissidents who disagree to "Old people are more reliable." and "40yo boss is more acceptable than 20yo one."

17 supporters/agists who agree to both statements.

70 inconsistent believers who agree to one age norm but disagree with the other.

Heterogeneity: dissidents of age norms

dissident = 1 if disagree to both statements

	Baseline	Base, Dissent	Spec2, Dissent
Leader=old	0.013	0.192	-0.110
	(0.23)	(0.28)	(0.40)
Culture=asian	-0.067	-0.029	-0.312
	(0.19)	(0.24)	(0.39)
Leader=old × Culture=asian	-0.513*	-0.763**	-0.395
	(0.31)	(0.37)	(0.47)
Leader=old × Culture=asian × Dissent		1.044	0.977
		(0.66)	(0.71)
Mean(depvar)	1.446	1.446	1.453
Prob > F	0.0345	0.0514	0.0866
N	121	121	106
Demographic controls	No	No	Yes
Heuristic controls	No	No	Yes

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Heterogeneity: supporters of age norms

support $= 1$ if agree to both statements

	Baseline	Base, Support	Spec2, Support
Leader=old	0.013	0.187	-0.052
	(0.23)	(0.26)	(0.32)
Culture=asian	-0.067	0.080	-0.092
	(0.19)	(0.21)	(0.24)
Leader=old × Culture=asian	-0.513*	-0.710**	-0.372
	(0.31)	(0.35)	(0.40)
Leader=old × Culture=asian × Support		1.260**	1.099
		(0.62)	(0.70)
Mean(depvar)	1.446	1.446	1.453
Prob > F	0.0345	0.1017	0.0181
N	121	121	106
Demographic controls	No	No	Yes
Heuristic controls	No	No	Yes

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Heterogeneity: inconsistent believers

consistent = 1 if agree to both or disagree to both

	Baseline	Base, Consistent	Spec2, Consistent
Leader=old	0.013	0.500	0.338
	(0.23)	(0.31)	(0.47)
Culture=asian	-0.067	0.225	0.161
	(0.19)	(0.28)	(0.43)
Leader=old × Culture=asian	-0.513*	-1.125***	-0.872
	(0.31)	(0.42)	(0.55)
Leader=old × Culture=asian × Consistent		1.496**	1.529**
		(0.60)	(0.69)
Mean(depvar)	1.446	1.446	1.453
Prob > F	0.0345	0.0516	0.0753
N	121	121	106
Demographic controls	No	No	Yes
Heuristic controls	No	No	Yes

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Your comments are welcome.

nhi.cao@hanken.fi

Thank you!

Power and Minimum Detectable Effect (MDE)

Number of groups per treatment:

- 5% significance level, 80% power
- $sd = 1.11$ (Curşeu et al., 2013)
- When groups are asked to solve 10 tasks in total, Bessey (2023) reports an MDE of 4 - treated groups solve 4 more tasks than comparison groups.

$\Rightarrow$ produces $N_{group} = 10$ groups within each treatment combination

No of players per group:

- Group size of 3-4 in Bessey (2023); Curşeu et al. (2013)

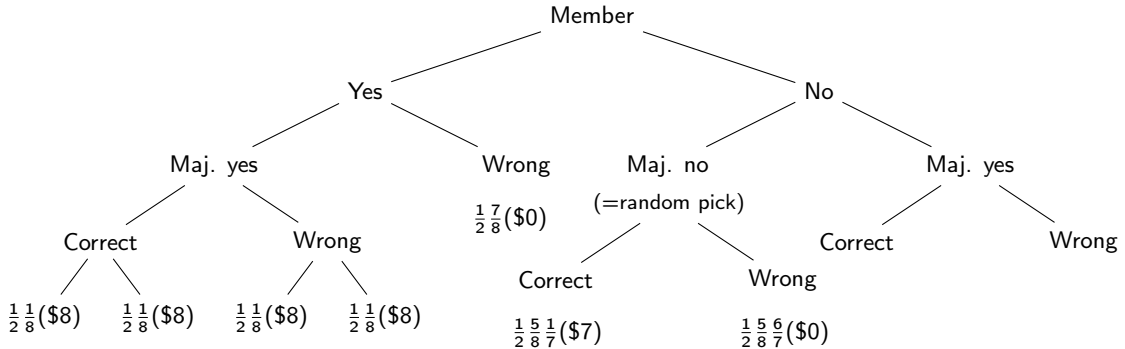
Total number of participants $N = 320$:

- $N_{participant} = N_{player} \times N_{group} \times N_{treatment} = 4 \times 10 \times 8 = 320$, treatment is any combination of $(Young, Old) \times (Asian, West, Mixed, Asian - anon.)$

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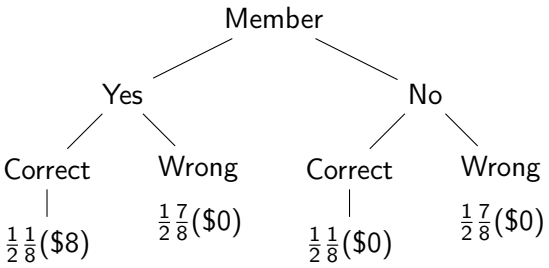
Incentive structure 1/

Prob(member votes Yes) $= \frac{1}{2}$ Prob(leader is correct) $= \frac{1}{8}$ Prob(majority votes No) $= 1 - (\frac{1}{2}\frac{1}{2} + \frac{1}{2}\frac{1}{2}\frac{1}{2}) = \frac{5}{8}$

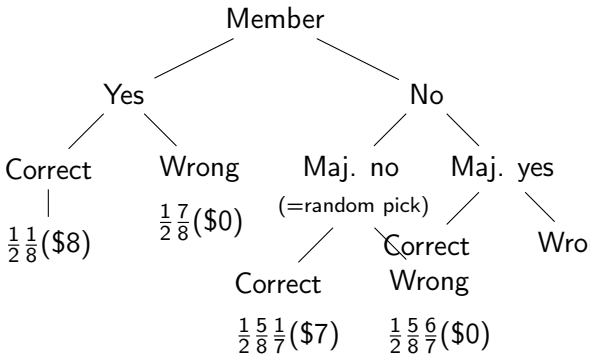


Incentive structure 2/

Prob(member votes Yes) $= \frac{1}{2}$ Prob(leader is correct) $= \frac{1}{8}$ Prob(majority votes No) $= 1 - (\frac{1}{2}\frac{1}{2} + \frac{1}{2}\frac{1}{2}\frac{1}{2}) = \frac{5}{8}$



$\$.5 > \$0$

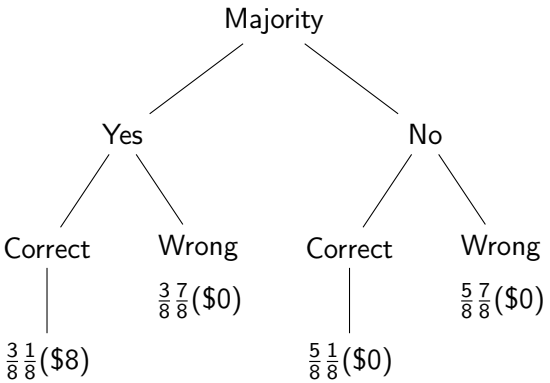


$\$.5 = \$.5$

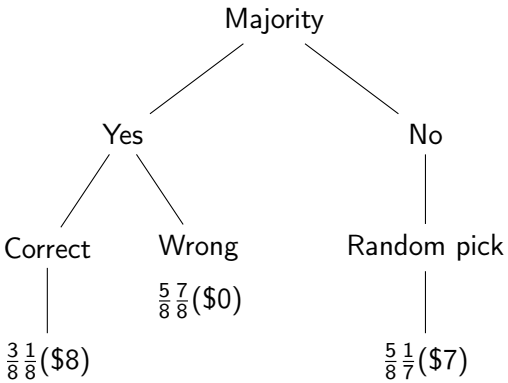
Incentive structure 3/

Prob(majority votes Yes) $= \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{8}$

Prob(leader is correct) $= \frac{1}{8}$



$.375 > $0



$.375 < $.625

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